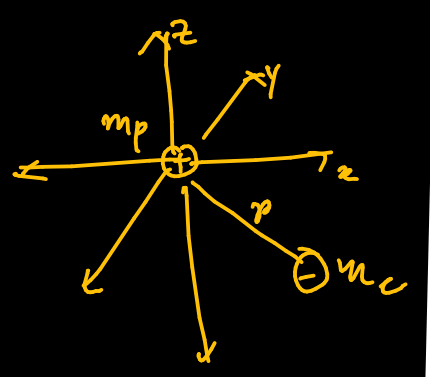


Atomic Orbital Nodes

$$\hat{H} = -\frac{\hbar^2}{2me} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r}$$

$$\hat{H}\Psi_{n,l,m}(r,\theta,\phi) = E_n \Psi_{n,l,m}(r,\theta,\phi)$$

$$\Psi_{n,l,m}(r,\theta,\phi) = R_{nl}(r) Y_l^m(\theta,\phi)$$



No. of Radial Nodes = $n - l - 1$

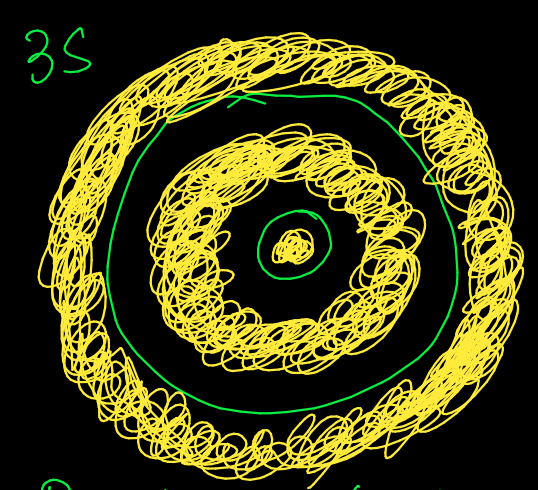
No. of angular Nodes = l

No. of Total Nodes = $(n - l - 1) + l = n - 1$

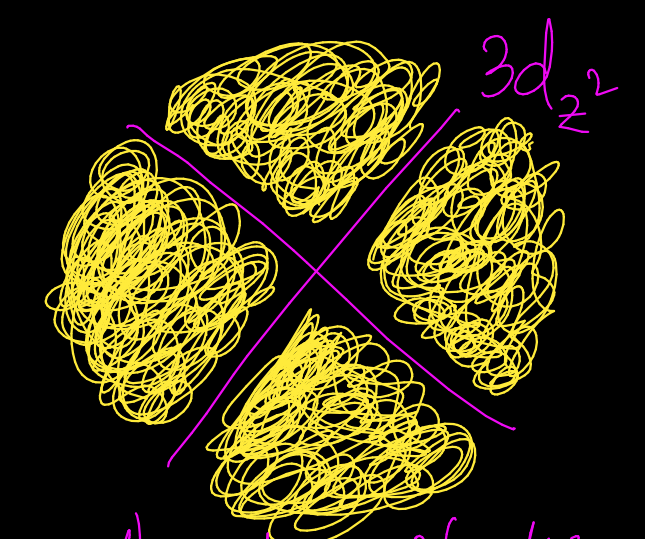
l	0	1	2	3	4	5	6	7	8	9	10	...
letter	s	p	d	f	g	h	i	k	l	m	n	...

$l=0$
 $l=0$
 $l=1$
 $l=0$
 $=1$
 $=2$
 $l=0$
 $=1$
 $=2$
 $=3$

Orbitals	N_{Ang}	N_{Rad}	N_{Total}
1s	0	0	0
2s	0	1	1
2p	1	0	1
3s	0	2	2
3p	1	1	2
3d	2	0	2
4s	0	3	3
4p	1	2	3
4d	2	1	3
4f	3	0	3



Radial Nodes



Angular Nodes

Radial wavefunction plots | Average value of radius of H-atom | Radial nodes | Angular nodes

Q: Find out average value of radius vector of 1s orbital for H-atom $\psi_{1s} = \frac{1}{\sqrt{\pi}} \frac{1}{a_0^{3/2}} e^{-r/a_0}$

$$\langle \hat{x} \rangle = \int \psi^* \hat{x} \psi d\tau$$

Vol^m element, $d\tau = \int_{r=0}^{\infty} r^2 dr \int_{\theta=0}^{\pi} \sin\theta d\theta \int_{\phi=0}^{2\pi} d\phi$

$$= 4\pi \int_{r=0}^{\infty} r^2 dr$$

Average value of radius:-

$$\langle r \rangle = 4\pi \int_{r=0}^{\infty} \psi^* \hat{r} \psi r^2 dr$$

$$= 4\pi \int_{r=0}^{\infty} \frac{1}{\sqrt{\pi}} \frac{1}{a_0^{3/2}} e^{-r/a_0} r \frac{1}{\sqrt{\pi}} \frac{1}{a_0^{3/2}} e^{-r/a_0} r^2 dr$$

$$= \frac{4\pi}{\pi a_0^3} \int_0^{\infty} r^3 e^{-2r/a_0} dr$$

$$= \frac{4}{a_0^3} \frac{\Gamma(3+1)}{\left(\frac{2}{a_0}\right)^{3+1}} \rightarrow \Gamma(4) = 3! = 6$$

$$= \frac{4}{a_0^3} \times \frac{a_0^4}{2^4} \times 6$$

$$\langle r \rangle = \frac{3}{2} a_0$$

Similarly, we can calculate

$$\langle r^2 \rangle = 3a_0^2 \quad \left\langle \frac{1}{r} \right\rangle = \frac{1}{a_0}$$

$$\left. \begin{aligned} \langle r \rangle &= \frac{3}{2} a_0 \\ \langle r^2 \rangle &= 3a_0^2 \\ \left\langle \frac{1}{r} \right\rangle &= \frac{1}{a_0} \end{aligned} \right\}$$

General Formula:

$$\langle r \rangle = \frac{1}{2} [3n^2 - l(l+1)] a_0$$

↑
principle quantum no.

↑
Angular quantum no.

1s

n=1
l=0

$$\langle r_{1s} \rangle = \frac{1}{2} [3(1)^2 - 0(0+1)] a_0 = \frac{3}{2} a_0$$

s p d f
l=0 1 2 3

$$\langle r_{2s} \rangle = \frac{1}{2} [3(2)^2 - 0(0+1)] a_0 = 6a_0$$

n=2 l=0

$$\langle r_{3d} \rangle = \frac{1}{2} [3(3)^2 - 2(2+1)] a_0 = \frac{21}{2} a_0$$

n=3
l=2

General formula for $\langle r^2 \rangle$, $\langle r^2 \rangle = \frac{n^2}{2} [5n^2 + 1 - 3l(l+1)] a_0^2$

1s
n=1
l=0

$$\langle r_{1s}^2 \rangle = 3a_0^2$$

Similarly we can calculate $\langle r_{2s}^2 \rangle$, $\langle r_{3d}^2 \rangle$, $\langle r_{2p}^2 \rangle$ -----

General formula for $\left\langle \frac{1}{r} \right\rangle = \frac{1}{n^2 a_0}$

1s
n=1
l=0

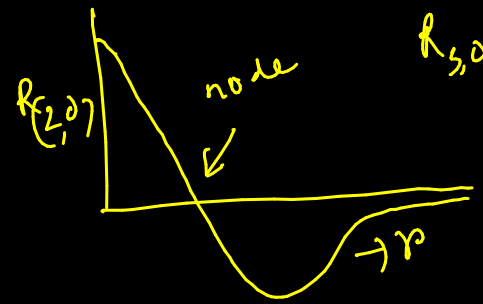
$$\left\langle \frac{1}{r_{1s}} \right\rangle = \frac{1}{a_0}$$

Wave Function Plots

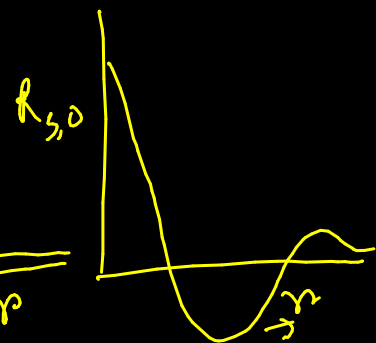
1s
 $n=1$
 $l=0$
 $\text{node} = n-l-1 = 0$



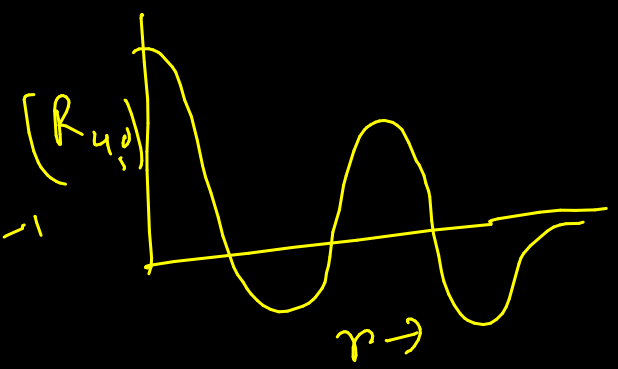
2s
 $n=2$
 $l=0$
 $\text{node} = n-l-1 = 2-0-1 = 1$



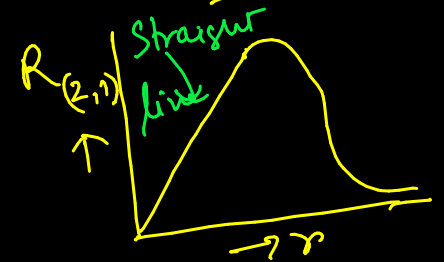
3s
 $n=3$
 $l=0$
 $\text{node} = 2$



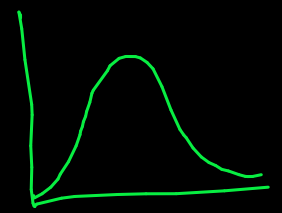
4s
 $n=4$
 $l=0$
 $\text{node} = n-l-1 = 3$



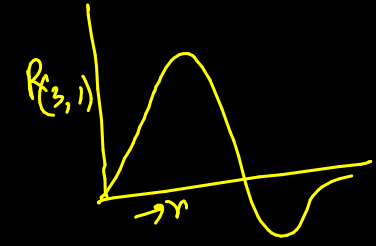
2p
 $n=2$
 $l=1$
 $\text{node} = n-l-1 = 2-1-1 = 0$



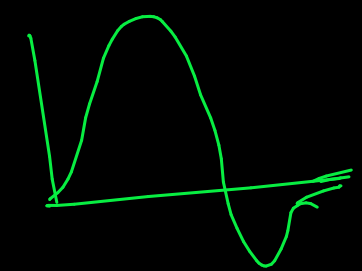
3d
 $n=3$
 $l=2$
 $\text{nodes} = 3-2-1 = 0$



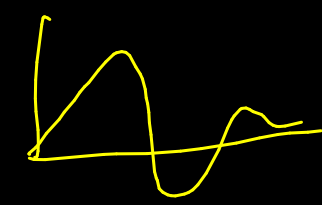
3p
 $n=3$
 $l=1$
 $\text{node} = 3-1-1 = 1$



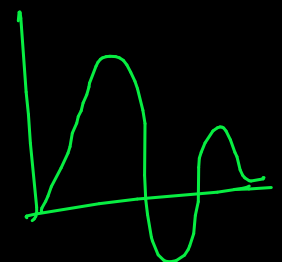
4d



4p
 $n=4$
 $l=1$
 $\text{node} = 4-1-1 = 2$



5d



Probability Density Plot

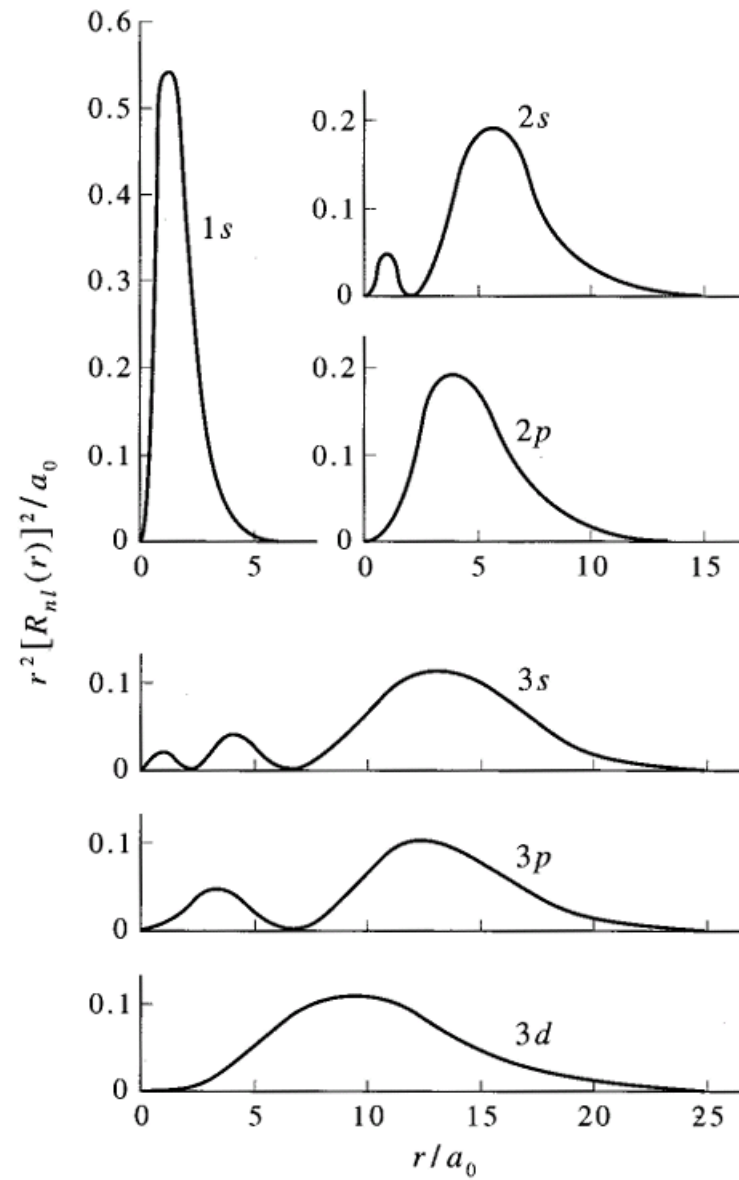


FIGURE 6.3

The probability densities $r^2[R_{nl}(r)]^2$ associated with the radial parts of the hydrogen atomic wave functions.

Hydrogen Wave Function

Probability density plots.

$$\psi_{nlm}(r, \vartheta, \varphi) = \sqrt{\left(\frac{2}{na_0}\right)^3 \frac{(n-l-1)!}{2n[(n+l)!]}} e^{-\rho/2} \rho^l L_{n-l-1}^{2l+1}(\rho) \cdot Y_{lm}(\vartheta, \varphi)$$

